Linear Algebra Solutions

Question 1: Find Orthogonal Complement & Prove w·w(tilda) = 0

Theory: Orthogonal Complement

The **orthogonal complement** of a subspace W, denoted as $W^\perp$ (or w(tilda)), is the set of all vectors that are orthogonal (perpendicular) to every vector in W.

Definition: $W^\perp = \{v \in V : v \cdot w = 0 \text{ for all } w \in W\}$

Key Properties:

1. $W^\perp$ is always a subspace
2. $\dim(W) + \dim(W^\perp) = n$ (dimension of the full space)
3. $(W^\perp)^\perp = W$
4. $W \cap W^\perp = \{0\}$

Problem Setup (3×3 Example)

Let's find the orthogonal complement of a subspace W in $\mathbb{R}^3$.

Given: W is spanned by vectors:

- $v_1 = [1, 2, 1]$
- $v_2 = [0, 1, 1]$

Step 1: Set up the orthogonality condition

A vector $\mathbf{x} = [x, y, z]$ is in $W^\perp$ if:

- $\mathbf{x} \cdot v_1 = 0$
- $\mathbf{x} \cdot v_2 = 0$

Step 2: Write the system of equations

From $v_1 \cdot \mathbf{x} = 0$: $x + 2y + z = 0$ From $v_2 \cdot \mathbf{x} = 0$: $y + z = 0$

Step 3: Solve the system

From equation 2: $z = -y$

Substitute into equation 1:
$x + 2y + (-y) = 0$

$x + y = 0$
$x = -y$

Step 4: Parametric solution

Let $y = t$ (parameter), then:

- $x = -t$
- $y = t$
- $z = -t$

x = t[-1, 1, -1]

Result: W⊥ = span{[-1, 1, -1]}

Proof that w · w(tilda) = 0

Theorem: If w ∈ W and u ∈ W⊥, then w · u = 0

Proof:

1. Let w be any vector in W. Then w can be written as: $w = c_1v_1 + c_2v_2 + ... + c_kv_k$ (linear combination of basis vectors of W)
2. Let u be any vector in W⊥. By definition of W⊥: $u \cdot v_i = 0$ for all basis vectors v_i of W
3. Now compute w · u: $w \cdot u = (c_1v_1 + c_2v_2 + ... + c_kv_k) \cdot u$
4. Using distributivity of dot product: $= c_1(v_1 \cdot u) + c_2(v_2 \cdot u) + ... + c_k(v_k \cdot u)$
5. Since u ∈ W⊥, each $v_i \cdot u = 0$: $= c_1(0) + c_2(0) + ... + c_k(0) = 0$

Therefore, w · w⊥ = 0 for all w ∈ W and w⊥ ∈ W⊥ ■

Verification for our example:

- $w = v_1 = [1, 2, 1]$
- w⊥ = [-1, 1, -1]
- w · w⊥ = (1)(-1) + (2)(1) + (1)(-1) = -1 + 2 - 1 = 0 ✓

Question 2: Gram-Schmidt Process

Theory: Gram-Schmidt Orthogonalization

The **Gram-Schmidt process** converts a set of linearly independent vectors into an orthogonal (or orthonormal) set that spans the same subspace.

Purpose: Create an orthogonal basis from any basis

Formula: For vectors $v_1, v_2, ..., v_n$:

- $u_1 = v_1$
- $u_2 = v_2 - \text{proj}(u_1)v_2 = v_2 - [(v_2 \cdot u_1)/(u_1 \cdot u_1)]u_1$
- $u_3 = v_3 - \text{proj}(u_1)v_3 - \text{proj}(u_2)v_3$
- And so on...

For orthonormal basis: normalize each u_i by dividing by its magnitude: $e_i = u_i/||u_i||$

Problem 1: 3×3 Matrix (3 vectors in ℝ³)

Given vectors:

- $v_1 = [1, 1, 0]$
- $v_2 = [1, 0, 1]$
- $v_3 = [0, 1, 1]$

Step 1: $u_1 = v_1$ $u_1 = [1, 1, 0]$

Step 2: Compute u_2

$\text{proj}(u_1)v_2 = [(v_2 \cdot u_1)/(u_1 \cdot u_1)]u_1$

$v_2 \cdot u_1 = (1)(1) + (0)(1) + (1)(0) = 1$
$u_1 \cdot u_1 = 1^2 + 1^2 + 0^2 = 2$

$\text{proj}(u_1)v_2 = (1/2)[1, 1, 0] = [1/2, 1/2, 0]$

$u_2 = v_2 - \text{proj}(u_1)v_2 = [1, 0, 1] - [1/2, 1/2, 0]$
$u_2 = [1/2, -1/2, 1]$

Step 3: Compute u_3

$\text{proj}(u_1)v_3 = [(v_3 \cdot u_1)/(u_1 \cdot u_1)]u_1$

$v_3 \cdot u_1 = (0)(1) + (1)(1) + (1)(0) = 1$
$\text{proj}(u_1)v_3 = (1/2)[1, 1, 0] = [1/2, 1/2, 0]$

$\text{proj}(u_2)v_3 = [(v_3 \cdot u_2)/(u_2 \cdot u_2)]u_2$

$v_3 \cdot u_2 = (0)(1/2) + (1)(-1/2) + (1)(1) = 1/2$
$u_2 \cdot u_2 = (1/2)^2 + (-1/2)^2 + 1^2 = 1/4 + 1/4 + 1 = 3/2$

$\text{proj}(u_2)v_3 = (1/2)/(3/2) \times [1/2, -1/2, 1] = (1/3)[1/2, -1/2, 1] = [1/6, -1/6, 1/3]$

$u_3 = v_3 - proj(u_1)v_3 - proj(u_2)v_3$
$u_3 = [0, 1, 1] - [1/2, 1/2, 0] - [1/6, -1/6, 1/3]$
$u_3 = [0 - 1/2 - 1/6, 1 - 1/2 + 1/6, 1 - 0 - 1/3]$
$u_3 = [-2/3, 2/3, 2/3]$

Orthogonal basis:

- $u_1 = [1, 1, 0]$
- $u_2 = [1/2, -1/2, 1]$
- $u_3 = [-2/3, 2/3, 2/3]$

To make orthonormal, normalize:

$e_1 = u_1/||u_1|| = [1, 1, 0]/\sqrt{2} = [1/\sqrt{2}, 1/\sqrt{2}, 0]$

$e_2 = u_2/||u_2|| = [1/2, -1/2, 1]/\sqrt{(3/2)} = [1/\sqrt{6}, -1/\sqrt{6}, 2/\sqrt{6}]$

$e_3 = u_3/||u_3|| = [-2/3, 2/3, 2/3]/\sqrt{(4/9+4/9+4/9)} = [-1/\sqrt{3}, 1/\sqrt{3}, 1/\sqrt{3}]$

Problem 2: 4×4 Matrix (4 vectors in $\mathbb{R}^4$)

Given vectors:

- $v_1 = [1, 0, 0, 0]$
- $v_2 = [1, 1, 0, 0]$
- $v_3 = [1, 1, 1, 0]$
- $v_4 = [1, 1, 1, 1]$

Step 1: $u_1 = v_1$ $u_1 = [1, 0, 0, 0]$

Step 2: Compute u_2

$v_2 \cdot u_1 = 1, u_1 \cdot u_1 = 1$
$proj(u_1)v_2 = 1 \times [1, 0, 0, 0] = [1, 0, 0, 0]$
$u_2 = [1, 1, 0, 0] - [1, 0, 0, 0] = [0, 1, 0, 0]$

Step 3: Compute u_3

$v_3 \cdot u_1 = 1, proj(u_1)v_3 = [1, 0, 0, 0]$
$v_3 \cdot u_2 = 1, u_2 \cdot u_2 = 1, proj(u_2)v_3 = [0, 1, 0, 0]$
$u_3 = [1, 1, 1, 0] - [1, 0, 0, 0] - [0, 1, 0, 0] = [0, 0, 1, 0]$

Step 4: Compute u_4

$v_4 \cdot u_1 = 1$, $\text{proj}(u_1)v_4 = [1, 0, 0, 0]$
$v_4 \cdot u_2 = 1$, $\text{proj}(u_2)v_4 = [0, 1, 0, 0]$
$v_4 \cdot u_3 = 1$, $\text{proj}(u_3)v_4 = [0, 0, 1, 0]$
$u_4 = [1, 1, 1, 1] - [1, 0, 0, 0] - [0, 1, 0, 0] - [0, 0, 1, 0] = [0, 0, 0, 1]$

Orthogonal basis:

- $u_1 = [1, 0, 0, 0]$
- $u_2 = [0, 1, 0, 0]$
- $u_3 = [0, 0, 1, 0]$
- $u_4 = [0, 0, 0, 1]$

Note: This particular set of vectors already produced the standard basis! They're already orthonormal.

Key Properties of Gram-Schmidt:

1. **Preserves span:** $\text{span}\{u_1, ..., u_k\} = \text{span}\{v_1, ..., v_k\}$ for all k
2. **Creates orthogonal set:** $u_i \cdot u_j = 0$ for $i \neq j$
3. **Works for any inner product space**
4. **Can be numerically unstable** (modified Gram-Schmidt is more stable)

Applications:

- QR decomposition
- Solving least squares problems
- Computing orthonormal bases for vector spaces
- Signal processing and Fourier analysis